

Undergraduate Curriculum in Agricultural Engineering

Land and Water Resources Engineering Option

2025-2026 Catalog

Total Credits 126

First Year (33 cr.)

Semester 1 (16 cr.)

R	Engr 1010	Orientation (FS)
3	A B E 1700	Graphics and Design (FS)
4	Chem 1670	General Chemistry (FS)
1	Chem 1670L	General Chemistry Lab (FS)
4	Math 1650	Calculus I (FS)
3	Engl 1500	Crit. Think. and Comm. (FSSS)
1	Lib 1600	Library Instruction (FSSS)

Semester 2 (17 cr.)

1	ABE 1100	Experiencing ABE (S)
3	ABE 1600	Engineering Problems (S)
4	Math 1660	Calculus II (FSSS)
4	Phys 2310	Classical Physics I (FSSS)
1	Phys 2310L	Classical Physics I Lab (FS)
3	Engl 2500	WOVE Composition (FSSS)
1	CE 2710	Engineering Foundations of Statics (FSSS)

Second Year (30 cr.)

Semester 3 (14 cr.)

3	A B E 2160	Fundamentals of AE (F)
2	CE 2720	Applied Engineering Statics (FS)
3	Math 2660	Differential Equations (FSSS)
3	Agron 1810	Intro to Crop Science (FS)
3	Agron 1820	Intro to Soil Science (FS)

Semester 4 (16 cr.)

2	ABE 2180	Project Mgmt & Design (S)
1	ABE 2010	Sophomore Seminar (FS)
3	ABE 3780	Mechanics of Fluids (FS)
1	ABE 3780L	Mechanics of Fluids Lab (FS)
3	ME 2310	Thermodynamics (FSSS)
3	Stat 3050	Engr Statistics (FSSS)
3	Biol 2510	Biological Processes in the Environment (S)
or	Biol 2110	Principles of Biology I (FS)

Third Year (32 cr.)

Semester 5 (16 cr.)

3	A B E 3160	Systems Modeling (FS)
4	A B E 3630	Electric Power and Electronics (FS)
3	CE 3720	Hydrology/Hydraulics (FS)
3	→	GIS Elective
3	→	Communication Elective

Semester 6 (16 cr.)

1	A B E 2730	CAD for Proc. Fac. And Land Use Planning
3	A B E 4310	Soil and Water Conservation (S)
3	EM 3240	Mechanics of Materials (FSSS)
2	Micro 2010	Introduction to Microbiology (FS)
1	Micro 2010L	Introduction to Microbiology Lab (FS)
3	→	ABE Breadth Elective
3	→	U.S. Cultures & Communities Elective

Fourth Year (31 cr.)

Semester 7 (17 cr.)

2	A B E 4150	AE Design I (FS)
3	A B E 4040	Instrumentation for Ag and Bio Engr (F)
3	Geol 2010	Geology for Engrs and Env Sci (F)
3	→	Water Quality Elective
3	→	International Perspectives Elective
3	→	SS/H Elective

Semester 8 (14 cr.)

2	ABE 4160	AE Design II (FS)
3	IE 3050	Engineering Economic Analysis (FSSS)
3	CE 3260	Principles of Environmental Engineering (FS)
3	→	Subsurface Systems Elective
3	→	SS/H Elective

***Highly recommended (not required) ***

FE Exam See adviser for review materials

***Please check the current catalog and Schedule of Classes for most recent offerings.**

Updated 9/3/25